

Total No. of Questions : 10]

SEAT No. :

PA-175

[Total No. of Pages : 2

[5927]-58

B.E. (Civil)

AIR POLLUTION AND CONTROL

(2015 Pattern) (Elective - III) (401009D) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Figures to the right indicates full marks.
- 3) Draw neat figures wherever necessary.

Q1) a) Explain the plume rise and how is it estimated? [5]

b) Enlist various zones of atmosphere. Explain with neat sketch and in detail: Troposphere and Ionosphere. [5]

OR

Q2) a) What is inversion? Briefly explain radiation and subsidence inversion?[5]

b) What are the effects of stack height? Enlist various formulae to calculate minimum stack height. [5]

Q3) a) Explain in detail with neat sketch working of a High Volume Sampler.[6]

b) What is sick building syndrome? Explain in detail. [4]

OR

Q4) a) Explain Radiation and Subsidence Inversion. [6]

b) Write methods of measurement of odour and explain any one method.[4]

Q5) a) Explain in detail: working principle and design of Fabric Filter as control equipment. [8]

b) Write advantages and disadvantages of [8]

i) Settling chamber

ii) Cyclone

OR

P.T.O.

- Q6)** a) Explain in detail working principle and design of electrostatic precipitator as a control equipment. [8]
b) Explain in brief [8]
i) Control of air pollution by process modification
ii) Control of air pollution by change of raw materials

- Q7)** a) How Land Use Planning is used as a method of control for air pollution? Explain in detail. [8]
b) Explain the important provisions made in Environment (Protection) Act 1986. [8]

OR

- Q8)** a) Enlist emission standards for stationary and mobile sources of air pollution. [8]
b) Explain in detail: Environment (Protection) Act 1986. [8]
Q9) a) Explain in detail: Role of regulatory agencies and control boards in obtaining environmental clearance for project. [9]
b) Explain in detail: Environmental Management Plan. [9]

OR

- Q10)** a) Explain in detail: environmental impact assessment (ETA) of a sugar industry. [9]
b) Explain in detail: the role of different authorities in EIA process. [9]

